

YUN ZHAO

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PROFESSIONAL SUMMARY

Senior Research Scientist with experience in Deep Learning research and application. Excited about GenAI research and real-world applications. Proficient in collaborating with high-performing teams to develop ML solutions to catapult business growth. Areas of expertise include Recommendation System, LLMs, and Deep Learning.

EXPERIENCE

Research Scientist, Monetization GenAI 04/2024-present
Meta Platforms, Inc. *Menlo Park, CA*

- Finetuned (SFT, RLHF) Llama to generate Ads' text, delivering 3% in segment Ads Score.

Research Scientist, Ads Core Machine Learning (Ads Ranking) 05/2021-03/2024
Meta Platforms, Inc. *Sunnyvale, CA*

- Created Ads ranking models by changing sub-models, placing features to the right modules, and hyperparameter tuning, to achieve global optima given any capacity, and technologies involved.
- Identified feature opportunities and develop high ROI features at scale.
- Developed feature interaction techniques (e.g., id matching, ngram, and bucketization) to improve feature learning.
- Selected the most promising, up-to-date features and generate a low-risk and high-quality feature baseline with the lowest latency possible.
- Enabled 2% Global Ads Score.

Software Engineer Machine Learning Internship 06/2020-09/2020
Facebook, Inc. *Menlo Park, CA*

- Deployed a creation time text selection model using BERT regression.
- Designed a linear ensemble model to combine BERT regression and other features to improve the performance in multi-text asset selection.
- Achieved a segment Ads Score lift of 1.44%.

EDUCATION

Ph.D. in Computer Science, University of California, Santa Barbara 09/2016-04/2021
Dissertation: Data Mining in Neuroscience and Healthcare.

M.E. in Electric Engineering, Tsinghua University 09/2012-06/2016
Thesis: Energy-efficient Resource Allocation in Mobile Cloud Computing.

B.E. in Electric Engineering, Zhejiang University 09/2008-06/2012
Thesis: Object Detection on RGB-D Images.

TECHNICAL SKILLS

- **Programming Software & Tools:** Python, C/C++, MATLAB, PostgreSQL, R, $\text{\LaTeX}$, PHP, HTML
- **Machine Learning & Statistics:** Predictive Modeling, Data Mining, Neural Networks
- **Deep Learning Frameworks:** Pytorch, Tensorflow, Caffe

SELECTED PUBLICATIONS

1. Yuqing Wang, **Yun Zhao**, “Metacognitive prompting improves understanding in large language models”, in NAACL, 2024.
2. Yuqing Wang, **Yun Zhao**, “Tram: Benchmarking temporal reasoning for large language models”, in Findings of ACL, 2024.
3. Yuqing Wang, **Yun Zhao**, “Gemini in Reasoning: Unveiling Commonsense in Multimodal Large Language Models”, in submission, 2024.
4. Yuqing Wang, **Yun Zhao**, and Linda Petzold, “Are Large Language Models Ready for Healthcare? A Comparative Study on Clinical Language Understanding”, in Machine Learning for Healthcare Conference, PMLR, 2023, New York, USA, Aug. 2023.
5. **Yun Zhao**, Qinghang Hong, Xinlu Zhang, Yu Deng, Yuqing Wang, Linda Petzold, “BERTSurv: BERT-Based Survival Models for Predicting Outcomes of Trauma Patients”, in ICDM 2021, New York, USA, Jul. 2021.
6. **Yun Zhao***, Yuqing Wang*, Junfeng Liu, Haotian Xia, Zhenni Xu, Qinghang Hong, Zhiyang Zhou, Linda Petzold, “Empirical Quantitative Analysis of COVID-19 Forecasting Models”, in DMBIH 2021, Auckland, New Zealand, Dec. 2021. (Recipient of the **Best Paper Award**)
7. **Yun Zhao***, Yuqing Wang*, Rachael Callcut, Linda Petzold, “Integrating Physiological Time Series and Clinical Notes with Transformer for Early Prediction of Sepsis”, in ICDM 2022, New York, USA, Jul. 2022.
8. **Yun Zhao***, Yuqing Wang*, Rachael Callcut, Linda Petzold, “Empirical Analysis of Machine Learning Configurations for Prediction of Multiple Organ Failure in Trauma Patients”, in ICDM 2021, New York, USA, Jul. 2021.
9. **Yun Zhao***, Xinlu Zhang*, Rachael Callcut, Linda Petzold, “Multiple Organ Failure Prediction with Classifier-Guided Generative Adversarial Imputation Networks”, in BIODDD 2021, Virtual, Aug. 2021.
10. **Yun Zhao**, Franklin Ly, Qinghang Hong, Tyler Santander, Henry T. Yang, Paul K. Hansma, Linda Petzold, “How Much Does It Hurt: A Deep Learning Framework for Chronic Pain Score Assessment”, in DMBIH 2020, Sorrento, Italy, Nov. 2020.
11. **Yun Zhao**, Richard Jiang, Zhenni Xu, Elmer Guzman, Paul K. Hansma, Linda Petzold, “Scalable Bayesian Functional Connectivity Inference for Multi-Electrode Array Recordings”, in BIODDD 2020, San Diego, CA, USA, Aug. 2020.
12. **Yun Zhao**, Elmer Guzman, Morgane Audouard, Zhuowei Cheng, Paul K. Hansma, Kenneth S. Kosik, and Linda Petzold, “A Deep Learning Framework for Classification of in vitro Multi-Electrode Array Recordings”, in ICDM 2019, New York, USA, Jul. 2019.
13. **Yun Zhao**, Sheng Zhou, Tianchu Zhao, Zhisheng Niu, “Energy-Efficient Task Offloading for Multiuser Mobile Cloud Computing”, in IEEE ICC 2015, Shenzhen, China, Nov. 2015.
14. Yuqing Wang, **Yun Zhao** and Linda Petzold, “Predicting the Need for Blood Transfusion in Intensive Care Units with Reinforcement Learning”, in ACM-BCB 2022, Chicago, IL, USA, Aug. 2022. (Recipient of the **Best Student Paper Award**)

HONORS AND AWARDS

Best Student Paper Award , 13th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics	2022
Best Paper Award , 9th Workshop on Data Mining in Biomedical Informatics and Healthcare	2021
Chancellor’s Fellowship , University of California, Santa Barbara (only one in CS Department)	2016
Holbrook Fellowship , University of California, Santa Barbara (annually awarded to 6 freshmen)	2016